CLOSURE PROPERTIES OF FPPF COVERINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [2155, 2168] and proof(s) [[2170, 2184]]; this spec covers the complete environment instead (statement 2155-2168, proof span(s) [(2170, 2184)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Editorial note. The proof is complete with cited stability facts; the one tail item (surjectivity of a base-changed family) is marked '(proof omitted)' by the author and is the standard set-theoretic bookkeeping.

Source reference. Original reference 01U7: sources/stacks-chow/tags.txt, source lines 2400–2400. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01TR: sources/stacks-chow/tags.txt, source lines 2384–2384. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01U9: sources/stacks-chow/tags.txt, source lines 2402–2402. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01TS: sources/stacks-chow/tags.txt, source lines 2385–2385. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let T be a scheme.*

(1) *If $T' \to T$ is an isomorphism then $\{T' \to T\}$ is an fppf covering of T.*
(2) *If $\{T_i \to T\}_{i \in I}$ is an fppf covering and for each i we have an fppf covering $\{T_{ij} \to T_i\}_{j \in J_i}$, then $\{T_{ij} \to T\}_{i \in I, j \in J_i}$ is an fppf covering.*
(3) *If $\{T_i \to T\}_{i \in I}$ is an fppf covering and $T' \to T$ is a morphism of schemes then $\{T' \times_T T_i \to T'\}_{i \in I}$ is an fppf covering.*

Proof. The first assertion is clear. The second follows as the composition of flat morphisms is flat (see Morphisms, Lemma 01U7) and the composition of morphisms of finite presentation is of finite presentation (see Morphisms, Lemma 01TR). The third follows as the base change of a flat morphism is flat (see Morphisms, Lemma 01U9) and the base change of a morphism of finite presentation is of finite presentation (see Morphisms, Lemma 01TS). Moreover, the base change of a surjective family of morphisms is surjective (proof omitted). □

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